

Statistical Report

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2026-06-07

Statistical Report on Fertility Data in Singapore

Abstract

Introduction

Can time series models trained on data from 1960–2012 accurately forecast Singapore’s continuing fertility decline over 2013–2024?

TO INCLUDE WHEN INTRODUCTION IS DONE

Include necessary abbreviations when writing it up

4. Justify SARIMA period 12 in Introduction (page 3) The periodogram doesn’t obviously point to period 12, so you need literature to justify it. Add a paragraph in the Introduction discussing the Chinese zodiac Dragon year effect on Singapore births. This is what makes the SARIMA(p,d,q)(P,D,Q)[12] models substantively motivated rather than just data-driven.

Methods

The modelling workflow was designed to answer our forecasting needs. The TFR and TLB series were extracted from the dataset, reshaped into annual time series from 1960–2024 and split into a training period from 1960–2012 and a testing period from 2013–2024.

Time plots and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests of the two series were used to check for trends, changes in variance and the transformations required before modelling. Both series showed evidence of non-stationarity and since ARIMA models require the time series to be stationary, transformation of the series was necessary prior to modelling.

The log transformation was used to stabilise variance and make the model proportional rather than absolute changes. Differencing was used to remove the trends shown inside the data.

Both series were then transformed using either just the log or the log with first or second differencing. These transformed series were then tested by the KPSS test to ensure an improvement of the stationarity of the series.

Two transformations showed potential as being stationary (first-differenced and logged, and second-differenced logged). With the KPSS test a small p-value shows evidence against stationarity. For TFR, the KPSS test gave that $p = 0.0406$ for the first-differenced logged series and $p = 0.1$ for the second-differenced logged series. For the logged TLB with both first and second differencing gave the same p-values of $p = 0.1$. So for TLB stationarity was not rejected for either transformation.

Despite the first difference of the TFR data having a KPSS test result below the significance threshold, we would prefer there to be less differencing as we do not want to over-difference the data and introduce noise in a relatively short data-set. We kept the second-differenced models as comparison as well so we could see another set of models in the results.

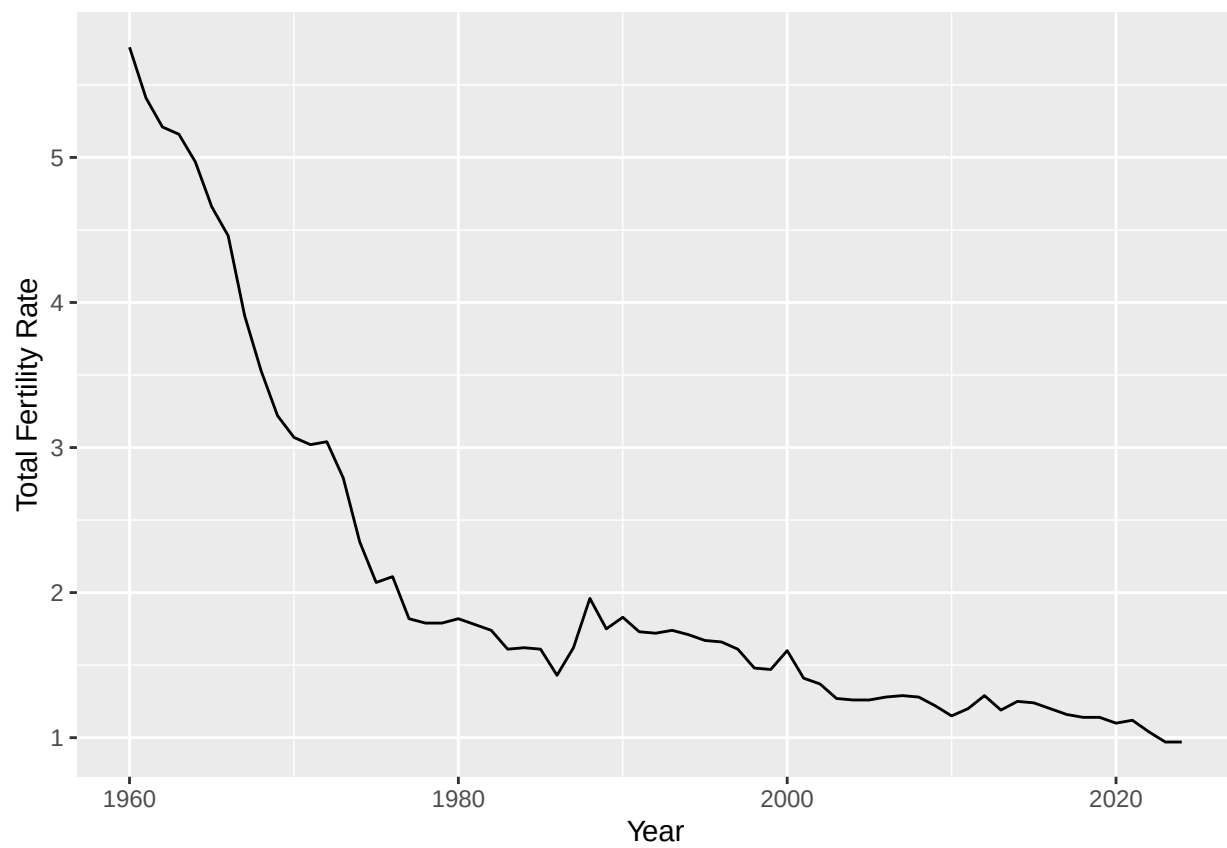


Figure 1: Annual Total Fertility Rate in Singapore

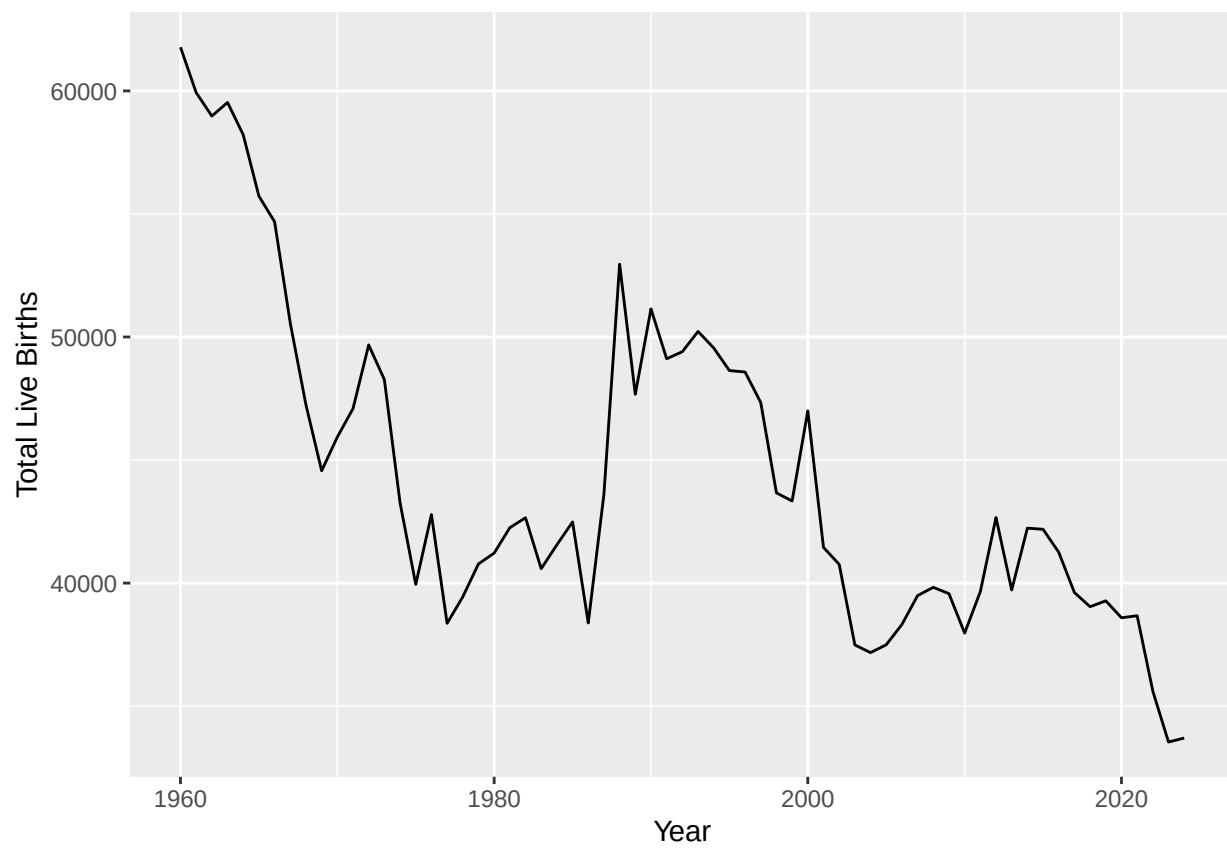


Figure 2: Total Live Births in Singapore

Because the transformed series and baseline residual diagnostics showed dependence around lags 11–13, higher-lag ARIMA models were also considered. These models were included to capture longer-lag autoregressive dependence that was not represented by the low-order baseline models.

SARIMA-style models with period 12 were also considered. Although the data are annual, the period-12 structure was motivated by both the lag-12 dependence observed in the transformed series and the literature discussed in the Introduction on Chinese zodiac-related birth timing effects in Singapore. These models were included to assess whether a 12-year pattern could be represented more simply than by unrestricted high-order ARIMA models. For the SARIMA candidates, the ACF and PACF plots of the transformed series were inspected to guide the initial period-12 model specification.

Models were only considered viable if their residuals were approximately white noise. This again was assessed with ACF and PACF, portmanteau tests and residual plots. Among viable models, the final model selection was based on AIC and forecast accuracy over the 2013-2024 period. Forecast accuracy was assessed with RMSE and MAE, with formulas provided in the appendix.

- So informed decision to try ARMIA (12,x, y) and SARIMA models potentially

Series	Models
TFR	SARIMA(0,1,1)(1,0,0)[12], SARIMA(1,1,1)(1,0,0)[12], SARIMA(1,2,1)(1,0,0)[12]
TLB	SARIMA(0,1,1)(1,0,0)[12], SARIMA(1,1,1)(1,0,0)[12], SARIMA(0,2,1)(1,0,0)[12]

Results

The ACF and PACF plots of the log-first-differenced series are shown in Figure 3. For both TFR and TLB the plots show a remaining dependence on lags 11-13, particularly at lag 12. This suggests that low order ARIMA models may not fully account for the higher order dependencies shown by Figure 3.

To assess the period-12 SARIMA candidates, the ACF and PACF of $(1 - B) \log(X_t)$ (Figure 3) and $(1 - B^{12})(1 - B) \log(X_t)$ (Figure 4) were used to guide the structure of our models. For both series, Figure 3 showed PACF spikes at lag-12 this suggested a seasonal AR term could be useful, while the lower lag ACF structure supported testing simple MA terms. This informed our decision for our first SARIMA candidate model to be SARIMA(0,1,1)(1,0,0)[12]. Nearby candidates were also fitted by varying the non-seasonal AR and MA orders and by comparing seasonal AR and MA terms.

In our seasonally differenced plot (Figure 4), lag-12 dependence remained even after the period 12 differencing. This indicated that differencing alone could not remove the period 12 dependence structure and supported the use of seasonal ARIMA models.

Firstly, we fitted some low-order ARIMA models without seasonal terms, however, as expected based on the lag-12 dependence identified with the ACF/PACF plots these models failed to produce white noise residuals. We next decided to fit some higher-order ARIMA models with seasonal terms as well as some SARIMA models with lag 12. You can see the tested models below with their resultant residuals:

Table 1: Ljung–Box residual tests for TFR and TLB candidate models

model_type	tfr_lb_pvalue	tfr_viable	tlb_lb_pvalue	tlb_viable
ARIMA(0,2,1)	0.005	No	0.003	No
ARIMA(1,2,1)	0.003	No	0.004	No
ARIMA(11,1,0)	0.000	No	0.001	No
ARIMA(12,1,0)	0.002	No	0.012	No
ARIMA(13,1,0)	0.020	No	0.045	No

model_type	tfr_lb_pvalue	tfr_viable	tlb_lb_pvalue	tlb_viable
ARIMA(13,1,1)	0.022	No	0.109	Yes
SARIMA(0,1,1)(0,0,1)[12]	0.027	No	0.028	No
SARIMA(0,1,1)(1,0,0)[12]	0.099	Yes	0.133	Yes
SARIMA(0,2,1)(1,0,0)[12]	0.004	No	0.049	No
SARIMA(1,1,1)(0,0,1)[12]	0.004	No	0.019	No
SARIMA(1,1,1)(1,0,0)[12]	0.108	Yes	0.115	Yes
SARIMA(1,2,1)(1,0,0)[12]	0.082	Yes	0.121	Yes

After testing each models residual autocorrelation we have a list of models which pass the test and can now evaluate their testing set errors to evaluate the model with the best prediction of both TFR and TLB in Singapore.

Table 2: AIC and AICc values for TFR candidate models

.model	AIC	AICc	BIC
tfr_sar_011_100_log	-137.096	-136.596	-131.243
tfr_sar_111_100_log	-136.479	-135.628	-128.674
tfr_sar_121_100_log	-132.258	-131.388	-124.531

Table 3: AIC and AICc values for TLB candidate models

.model	AIC	AICc	BIC
tlb_sar_011_100_log	-146.736	-146.236	-140.882
tlb_sar_111_100_log	-145.079	-144.228	-137.274
tlb_sar_121_100_log	-138.646	-137.776	-130.918
tlb_ar13ma1_log	-141.446	-128.113	-112.178

Table 4: Forecast accuracy for viable TFR models over 2013–2024

.model	RMSE	MAE	MAPE
tfr_sar_121_100_log	0.063	0.049	4.478
tfr_sar_011_100_log	0.098	0.073	6.945
tfr_sar_111_100_log	0.132	0.102	9.671

Table 5: Forecast accuracy for viable TLB models over 2013–2024

.model	RMSE	MAE	MAPE
tlb_sar_121_100_log	3035.560	2280.238	6.151
tlb_sar_011_100_log	3667.483	2752.134	7.542
tlb_sar_111_100_log	3914.953	2943.136	8.091
tlb_ar13ma1_log	10016.593	8587.467	23.304

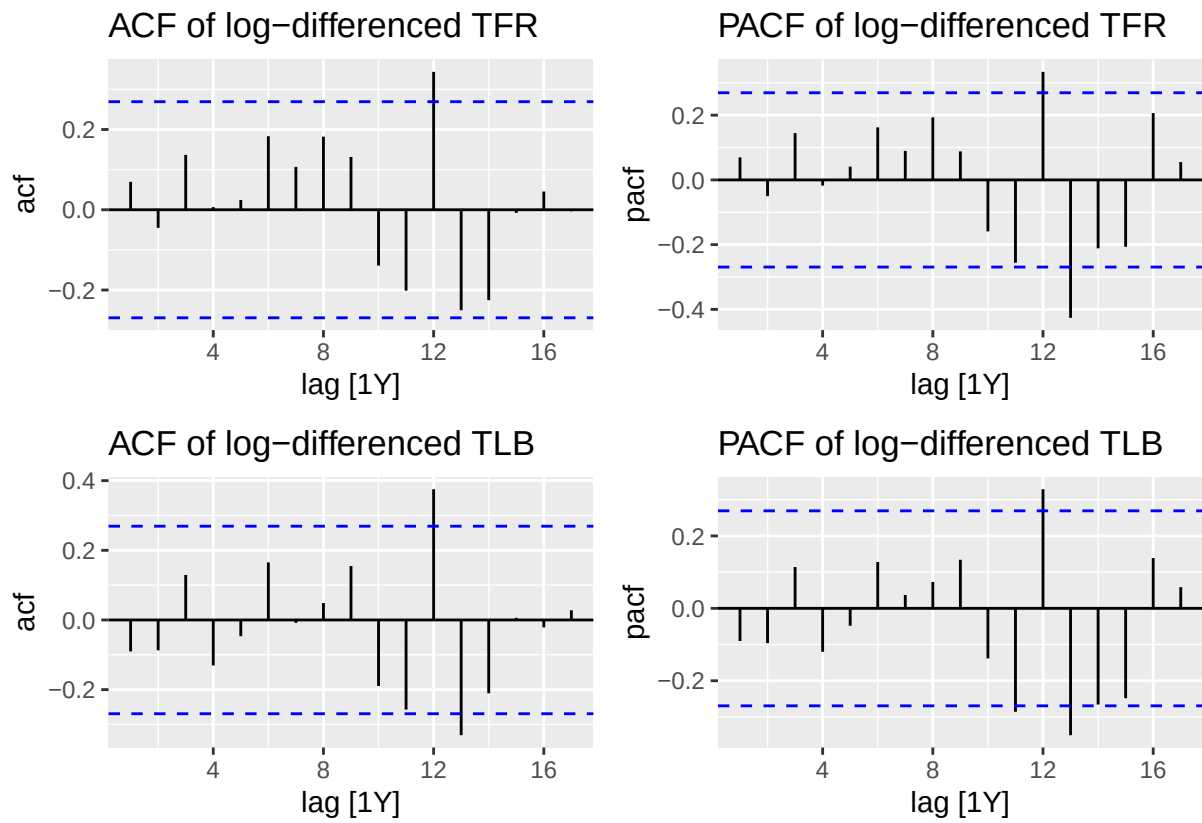


Figure 3: ACF and PACF plots of the log-first-differenced TFR and TLB series

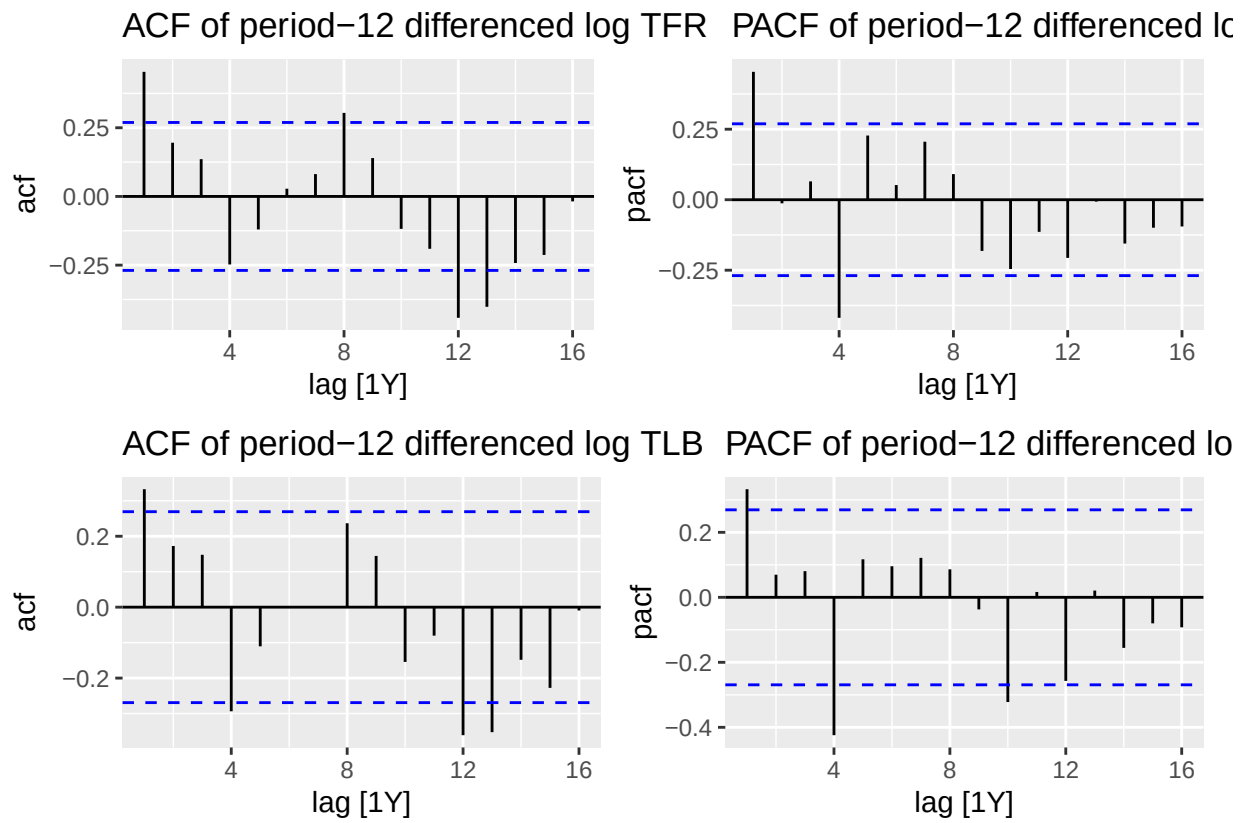
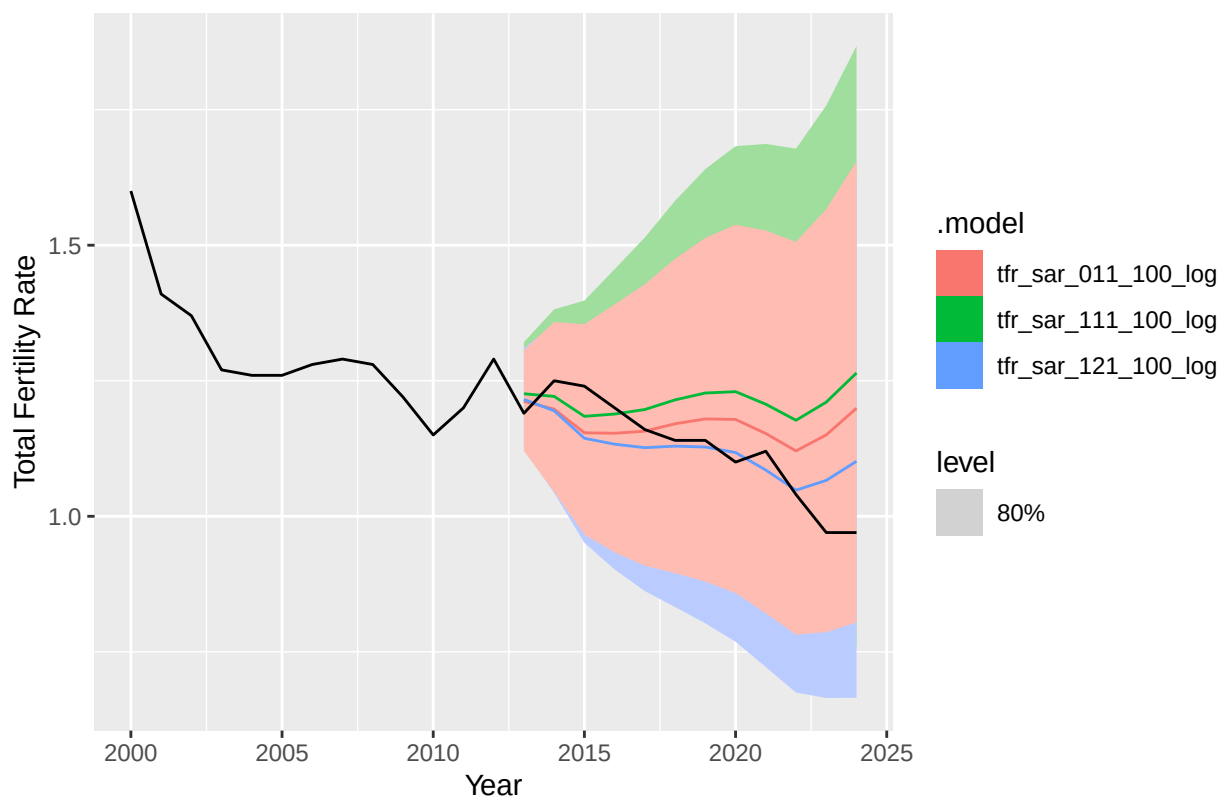
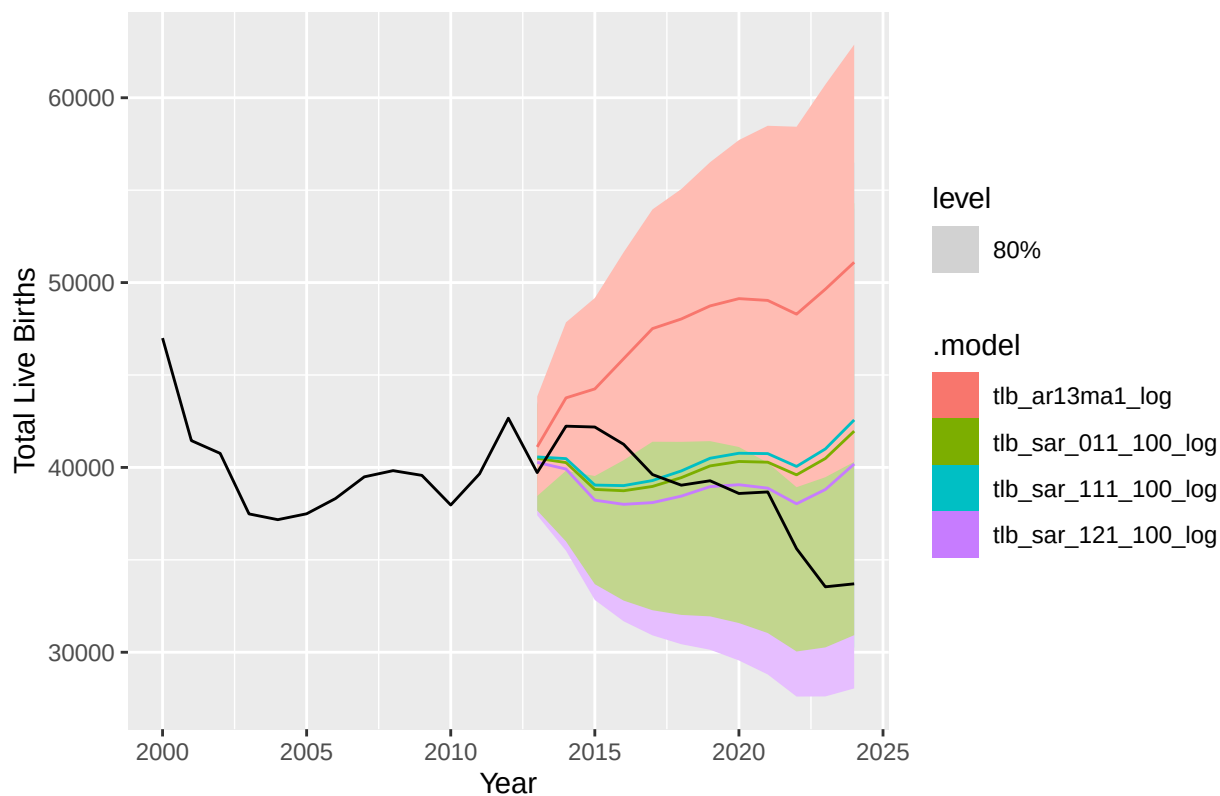


Figure 4: ACF and PACF plots of the log-first-differenced and period-12 differenced TFR and TLB series

TFR: Viable SARIMA forecasts vs observed (2000–2024)



TLB: Viable model forecasts vs observed (2000–2024)



Your professor expects the Results section to show:

ACF/PACF plots of the detrended and deseasonalised series
How the first viable model was identified from those plots
How you varied parameters to get additional viable models
How each viable model fits the testing data

Discussion

Conclusion

References

Appendices if applicable.